

Juan David Muñoz-Bolaños

Optical Metrology Engineer

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PROFESSIONAL SUMMARY

Optical Metrology Engineer with a specialized background in high-precision metrology, optical simulation, and Python/C++ automation. Proven expertise in building automated optical test setups from scratch and performing rigorous mathematical data analysis to guide technical design and system integration.

TECHNICAL COMPETENCIES

Optical Metrology & Simulation

Expert in wavefront sensing, phase retrieval, and ZEMAX OpticStudio tolerancing. Proficient in characterizing aberration sensitivity of high-NA microscope objectives.

System Automation & Control

Experience building C++/Python/LabVIEW control stacks for adaptive optics, scientific cameras (sCMOS), and opto-mechanical stages.

Data Science for Optics

Application of supervised ML and AI algorithms to hyperspectral imaging data. Proficient with NumPy, Pandas, Scikit-learn, and JAX for high-performance computation.

Hardware Integration & Sourcing

End-to-end R&D experience in multi-photon microscope assembly. Competent in technical component specification (Thorlabs, Edmund Optics) and incoming QA.

Process Reliability

Hands-on experience with industrial vision solutions (Halcon) and ensuring measurement traceability via Gage R&R assessments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **Automated Metrology Setups:** Designed and built an automated wavefront sensing system; developed the full software control stack to unify scientific cameras and wavefront modulators.
- **Optical Modeling & Design:** Performed systematic ZEMAX tolerance analysis to optimize the process of custom objective lens characterization and aberrometry.
- **Integration & Alignment:** Achieved diffraction-limited performance in a custom-built nonlinear microscope by precisely aligning galvo scanners, femtosecond lasers, and PMT detectors.
- **Technical Documentation:** Managed technical specification sheets and qualified vendors (Hamamatsu, Thorlabs) for high-precision opto-mechanical integration.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Spectrometer Build:** Handled the end-to-end optical layout and detector calibration of a 1064 nm fiber-probe Raman spectrometer prototype.
- **AI-Driven Data Analysis:** Leveraged custom Python packages for Machine Learning classification on hyperspectral tissue images, accelerating experimental turnaround.
- **Published Research:** Contributed to the development of non-invasive clinical diagnostic tools, results documented in peer-reviewed journals (Applied Sciences, ACS Photonics).

INTECOL S.A.S.

Medellín, Colombia

- **Industrial Machine Vision:** Engineered Halcon inspection systems for beverage production lines; solved critical label and sealing quality issues via custom illumination design.
- **Hardware Commissioning:** Optimized robotic vision for auto-part recognition using Cognex VisionPro, directly impacting production efficiency at client sites like Renault.
- **Gage R&R Assessments:** Verified system accuracy and process stability within industrial quality control frameworks.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Microscopy

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Head of Institute for Biomedical Physics

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Medical University of Innsbruck

Dr. Christoph Krafft

Group leader: Spectroscopy Imaging

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Leibniz Institute of Photonic Technology